Codebook

An AI teammate that remembers how each partner thinks.

Team Rubixx · Rajendra Sarpal

DEMO VIDEO youtu.be/eg9yptg7180

LIVE DEMO codebook-755276882523.asia-south1.run.app

CODE github.com/Rajendra465/graph-hack-sub

AI assistants forget the people they work with.

- ✕ Every session starts from zero.
- ✕ You get the same answers as everyone else.
- ✕ Hand work to another agent, and what it learned about you is lost.

✓ People don't work that way: good partners build a shorthand. Say **ocean**, and they think **wave**, not **bank**.

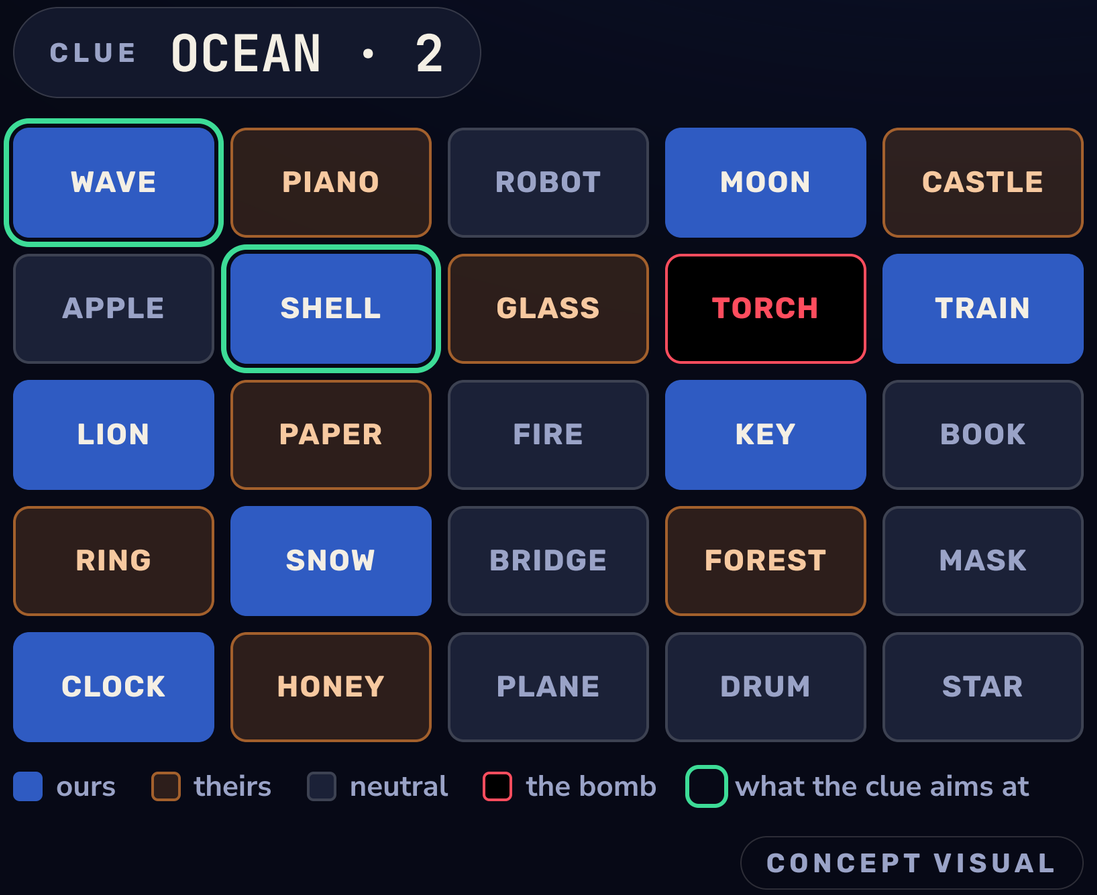
Can an agent keep a separate memory of each person it works with, and does that memory measurably help?

To measure it, we needed a task where knowing your partner can be scored.
A word game is exactly that.

Codenames in 20 seconds

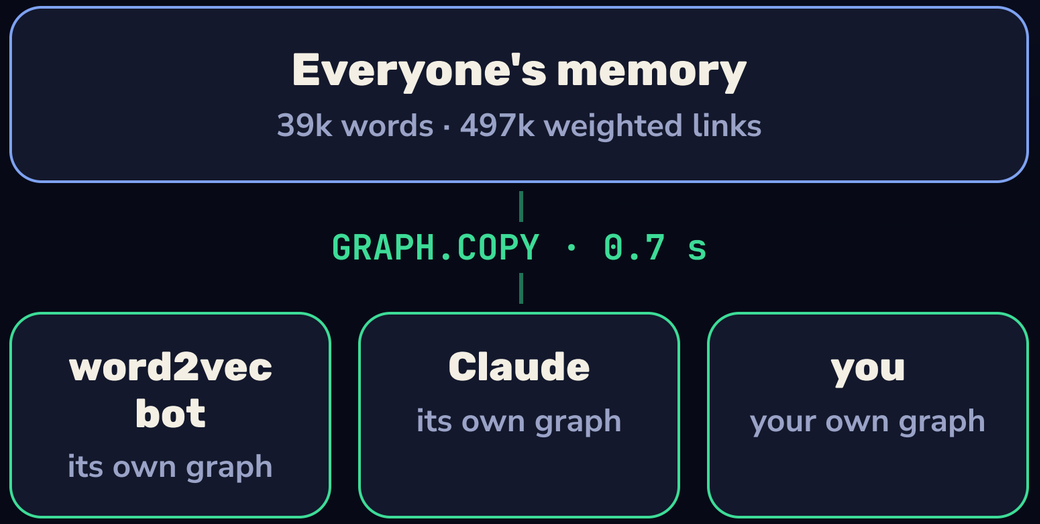
1. 25 word cards. Our AI, the **clue-giver**, knows which are ours.
2. It says one word and a number: `OCEAN · 2`.
3. Its **partner** sees only the words, and guesses.
4. One card is **the bomb**. Touch it, and you lose.

Good clues depend on how this partner links words: that is what the memory learns.



One shared memory, plus a private copy per partner.

- A **memory** is a graph: words linked by how strongly one brings the other to mind.
- Everyone's memory: **39k words and 497k links** in **FalkorDB**, a graph database.
- Each partner gets **a copy in 0.7 s**. Every guess makes its links stronger or weaker.



CONCEPT VISUAL

Every turn: find, score, block, learn.

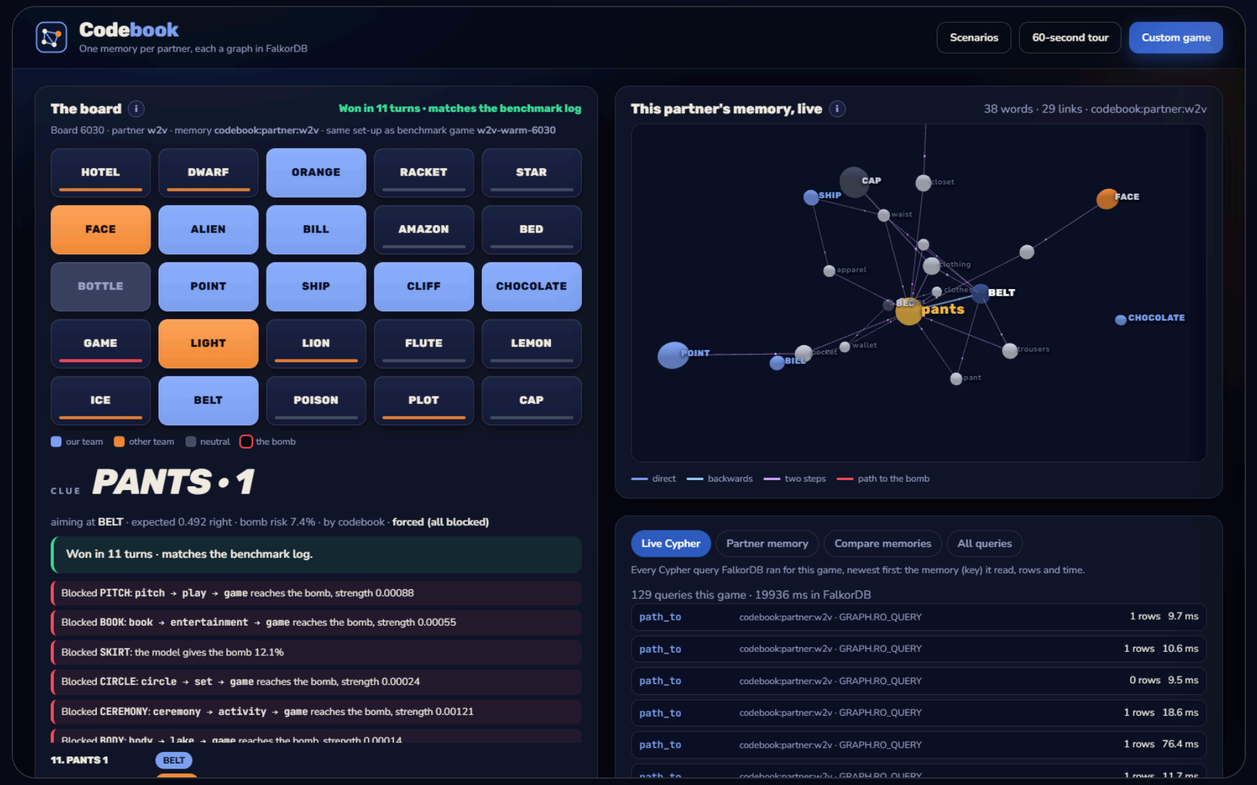
Find Cypher, the graph query language, reads 4 kinds of link from each clue to the board.

Score Simulate this partner's picks; rank the clues.

Block Refuse any clue whose chain of links reaches the bomb.

Learn Re-weight this partner's links after each guess.

The live view, board 6030: the clue, a plain-English narration and the partner's memory in 3D



06 · ARCHITECTURE

Any agent plugs in. FalkorDB holds every memory.

Any AI agent
6 MCP tools (MCP is the standard way AI agents call tools)

Command line
codebook setup · train · play · eval · ui

Live view
every Cypher query as it runs, and the 3D graph

↓

Codebook engine scores every clue · blocks risky ones · learns from each guess

↓

FalkorDB

`codebook:population`
everyone's memory, read-only

`codebook:partner:<id>`
one per partner, each locked to its own database user

A memory per person is built in.

1 graph
per partner, each its own key

0.7 s
GRAPH.COPY makes a new partner's memory

1 user
per partner: database ACLs lock each graph to its owner

Milliseconds
Cypher traversals, every turn

11 MB
DUMP/RESTORE carries a memory to another server

Restart-proof
the same games replay with the same fingerprint

Remove FalkorDB and all 13 commands fail: there is no fallback store.

3 of 4 promises kept, all written before testing.

BOMB HITS

32% → 21%

word2vec bot, 200 boards it never saw: its own memory vs everyone's

TURNS TO FINISH

15.55 → 14.39

lower is better; a lost game counts 25

The more games together, the better it plays

word2vec bot · turns to finish on the same 200 boards

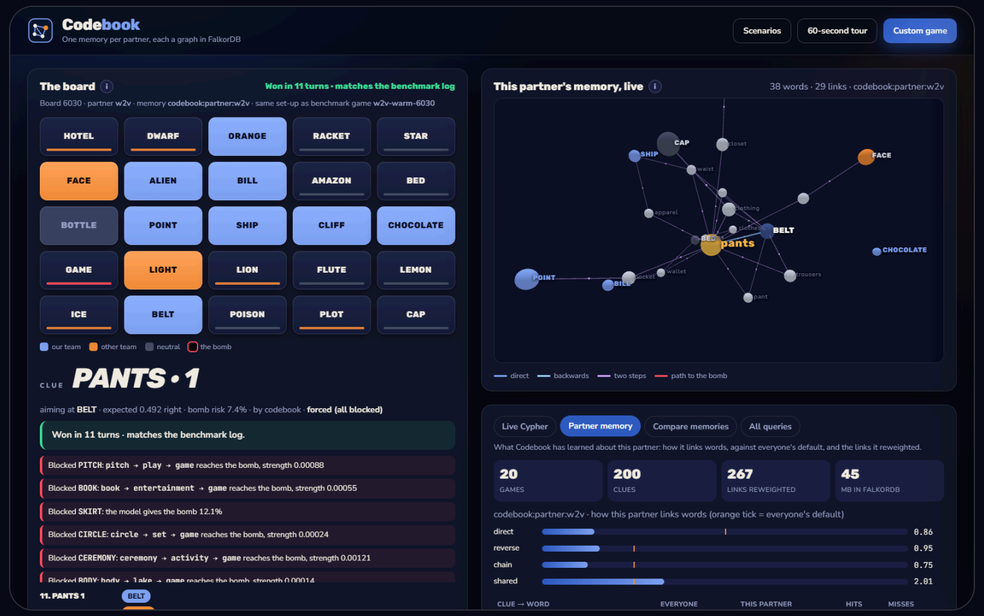


Not significant, and said plainly: the GloVe bot, and the handover to Claude Sonnet (bomb hits 40% → 22%, but only 40 boards).

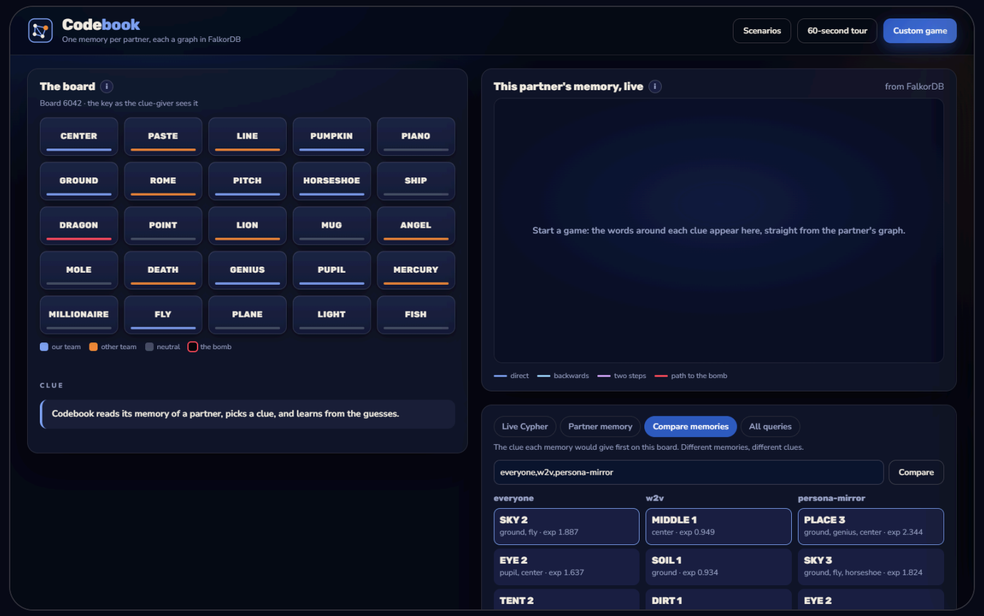
Another AI can use the memory, safely.

Claude gives clues from the bot's memory through MCP tools, with no database password; risky clues are still blocked. You can play too, and your own memory appears.

What one partner's memory holds



Same board, three memories, three clues



Every agent that works with people needs a memory of each person.

Coding assistant

learns your team's shorthand

its own memory graph

Support agent

remembers each customer

its own memory graph

Tutor

adapts to every student

its own memory graph

EXAMPLES, NOT SHIPPED INTEGRATIONS

Try it yourself.

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```
git clone https://github.com/Rajendra465/graph-hack-sub.git && cd graph-hack-sub
docker compose up -d                # FalkorDB 6.0.2
uv sync --extra dev
uv run codebook setup --source conceptnet --vectors
uv run codebook ui                  # http://127.0.0.1:7811
```

Limits, plainly: everyone's memory is built from ConceptNet 5.7, public word-association data. The partner model is small: 20 games find a habit, not every favourite word.